Project Report

Rishta System – A Digital Matrimonial Platform

Submitted by

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Rishta System – A Digital Matrimonial Platform

A project submitted to the

Department of Business Administration

In

Partial Fulfilment of the Requirements for the

Bachelor's Degree in Information Systems

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This is to certify that the project titled "Rishta System – A Digital Matrimonial Platform" is the genuine work carried out by Haseeb Javed , student of BBIS of Business Administration Department, University of Sahiwal, Sahiwal during the academic year [Your Session] , in partial fulfilment of the requirements for the award of the Bachelor of Information Systems degree, and that the project has not previously formed the basis for the award of any other degree, diploma, fellowship, or any other similar title..

Haseeb Javed___________________________

DECLARATION

This is to declare that the project entitled "Rishta System – A Digital Matrimonial Platform" is an original work done by undersigned, in partial fulfilment of the requirements for the degree "Bachelors of Information Systems" at Business Administration Department, University of Sahiwal, Sahiwal.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

Haseeb Javed______________________________

ACKNOWLEDGEMENTS

First and foremost, we are grateful to Allah Almighty, who blessed us with the health, ability, and guidance to complete this project successfully. Nothing could have been achieved without His help.

We express our sincere gratitude to our loving parents and teachers whose support enabled us to reach this stage. We also extend our appreciation to our Head of Department, Dr. Waris Ali, and our Coordinator, Dr. Atif, for their valuable support and guidance throughout our studies.

Finally, we would like to extend special thanks to Umair Waqas for providing us with an excellent learning experience during this project. His constructive feedback inspired us to explore new ideas and techniques for creating a secure and efficient digital matrimonial platform.

DEDICATION

We would like to dedicate this accomplishment to our beloved and caring Parents, friends, and to our teachers, with the support of whom I am standing at this step of my life stairs.

We also dedicate this dissertation to Umair Waqas who has supported us throughout the process

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LIST OF ABBREVIATION

ORS	Online Rishta System
UI	User Interface
UX	User Experience
DB	Database
CTA	Call to Action
FAQ	Frequently Asked Questions
KYC	Know Your Customer
CMS	Content Management System
PDF	Portable Document Format
CSV	Comma-Separated Values
ID	Identification
SSL	Secure Sockets Layer

Abstract

The "Rishta System" is a comprehensive digital matrimonial platform designed to connect individuals seeking life partners in a secure, efficient, and culturally appropriate manner. The core idea is to create an online ecosystem that simplifies the matchmaking process, moving beyond traditional methods that are often time-consuming and limited in reach. This project addresses key challenges such as privacy concerns, profile authenticity, and effective communication between potential matches. The system is divided into two main interfaces: a user-facing side and an administrative backend. The user side allows individuals to register, create detailed profiles with privacy controls, search for matches using advanced filters, send interest requests, and communicate via a chat system that unlocks only upon mutual acceptance. Key features include profile photo watermarking for security, a compatibility score to guide users, and a subscription model for premium access. The admin side provides a powerful dashboard for managing users, verifying profiles, monitoring messages for inappropriate content, handling subscriptions, and generating reports. The system is built using HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL, with XAMPP providing the local server environment. This combination ensures a robust, scalable, and easily deployable web application. The system aims to provide a trusted, user-friendly, and feature-rich platform that respects cultural values while leveraging modern technology to help people find their perfect match.

Chapter 1

Introduction

In today's fast-paced world, the traditional methods of finding a life partner—such as relying on family networks, word-of-mouth, or newspaper advertisements—have become less effective and more time-consuming. These methods offer limited reach, lack proper privacy, and provide no standardized way to filter potential matches based on personal preferences. Many individuals and families face difficulties in finding suitable matches due to geographical, cultural, or informational barriers. The digital transformation has touched almost every aspect of life, yet the matchmaking process in many societies remains largely offline and inefficient.

This project is conceived to address these limitations by providing a digital platform where users can create profiles, define their preferences, and connect with others in a controlled and respectful manner. The "Rishta System" leverages modern web technologies to offer a secure, scalable, and user-friendly solution. It empowers users to take control of their search, ensures authenticity through admin verification, and fosters meaningful connections through a safe communication channel.

The system is built using open-source technologies: HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL, with XAMPP as the local development environment. This stack was chosen because of its stability, wide community support, and ease of deployment. The platform is designed to be responsive, ensuring accessibility across desktops, tablets, and mobile devices.

1.1 Tools and technology used

The development of the Rishta System utilizes a combination of front-end, back-end, and database technologies to deliver a complete web-based solution. Below is a detailed breakdown:

- HTML5
- CSS3
- Bootstrap 4/5
- JavaScript (ES6)
- PHP 7.4+
- MySQL
- XAMPP

1.1.1 External Interface Requirements

1. **User Interface**

The user interface is built with Bootstrap to ensure responsiveness across all devices. It includes intuitive navigation, form elements, modals, and alerts. All forms are validated both client-side (JavaScript) and server-side (PHP).GUI along with meaningful frames and button

2. Hardware Interface

The system is designed to run on standard hardware used in academic and professional environments.

Hardware Environment	Intel Core i5
System Configuration	8 GB (minimum 4 GB)
Operating system	Windows 10 / 11, macOS, or Linux

Table 1 Hardware Interface

3. Software Interface

Front End	HTML5, CSS3, Bootstrap 5, JavaScript

Back End	PHP (7.4 or higher)

Table 2 Software Interface

Summary of Requirements ((SR) Initial Requirements)

1.1.2 User Account Management

1. Users must register with full name, email, password, gender, and date of birth.
2. Email must be unique and valid.
3. Minimum age: 18 years (calculated from date of birth).
4. Passwords must be at least 8 characters with mixed case (and optionally numbers).
5. After registration, user can log in using email and password.
6. Session management: user remains logged in until logout or session expiry.

1.1.3 Profile Management

1. Each user can create and edit a profile with fields: city, education, profession, religion, bio, and phone (optional).
2. Profile photos can be uploaded (multiple photos, one primary).
3. Photo privacy: users can choose who sees each photo (public, matched only, premium only).
4. Profile photos are automatically watermarked on upload using PHP GD library to prevent misuse.
5. Profile completion percentage is shown to encourage users to complete details

1.1.4 Search and Matching

1. Users can search profiles using filters: age range, city, education, religion, profession.
2. Results can be sorted by last active, profile completeness, or premium membership.
3. Compatibility score (0-100%) is calculated based on matching preferences and displayed.
4. Suggested matches are shown on dashboard based on user's profile and interests.

1.1.5 Interest Requests and Communication

1. Users can send "interest" requests to other profiles.
2. Interest status can be pending, accepted, or rejected.
3. Chat is only unlocked after mutual acceptance of interest.
4. Hidden information (phone, email) is revealed only after mutual acceptance.
5. Each message can be reported by the recipient; flagged messages are sent to admin

1.1.6 Record Management (Admin)

1. Admin has full CRUD access to user accounts (view, edit, suspend, delete).
2. Admin can reset user passwords.
3. Admin can view all interests, messages, and reports.
4. Admin can generate reports (user lists, activity logs, revenue summaries).

1.2.6 Subscription Management

1. Two subscription tiers: Free and Premium.
2. Free users have limited profile views per day (e.g., 10), blurred photos for other profiles, and chat disabled.
3. Premium users have unlimited views, full photo visibility, chat enabled, and priority in search results.

Gantt Chart

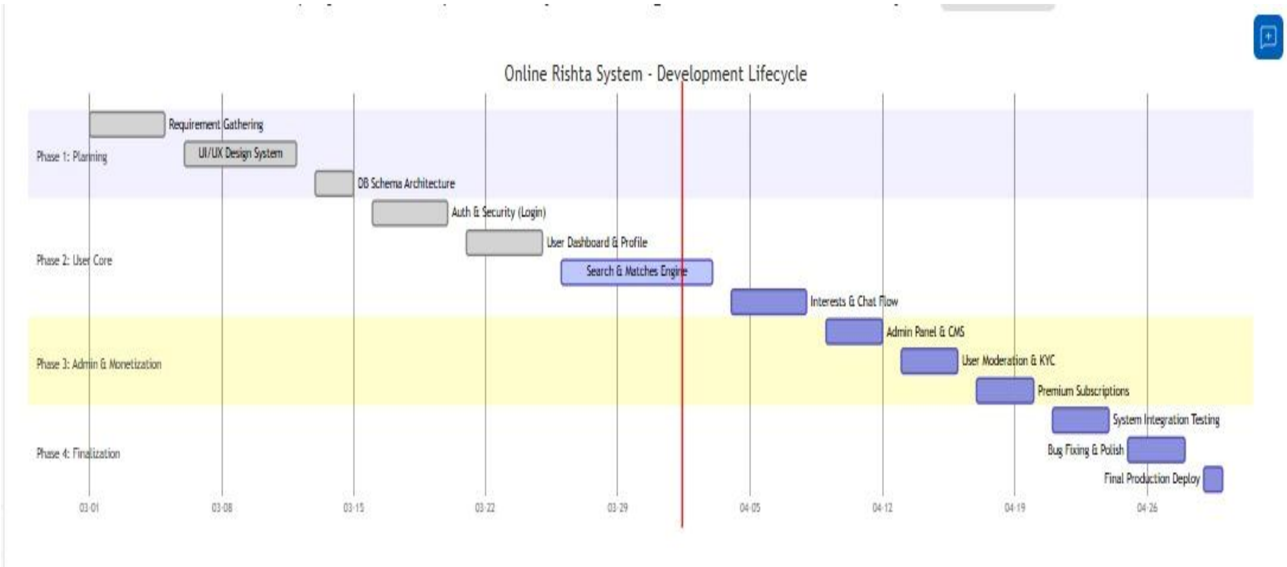

Figure 1 Gantt chat

1.2 Project / Product Feasibility Report (PFR)

A project idea hands down several insights for an in-depth investigation to further analyse the project. Hence there is a need to conduct a feasibility study by exploring many aspects of the project. A feasibility report has been prepared for Online Campus Depot Android application, addressing the following feasibilities:

1. Technical
2. Operational
3. Economic
4. Schedule
5. Specification
6. Information
7. Motivational
8. Legal and Ethical

1.2.1 Technical Feasibility

The project uses mature, open-source technologies (HTML, CSS, Bootstrap, JavaScript, PHP, MySQL). These technologies are well-documented, and there is ample community support. Our team has the required skills to develop the system using these tools. The XAMPP environment allows for seamless local development and testing. Therefore, the project is technically feasible.

1.2.2 Operational Feasibility

The system is designed with a user-friendly interface. Both tech-savvy and non-technical users can navigate easily. The admin panel provides comprehensive tools to manage the platform with minimal training. Automated processes (matching, interest handling) reduce manual effort, making the system operationally efficient.

Economic Feasibility

The development cost is minimal as all tools are free and open-source. The main cost would be for hosting and domain if deployed live. However, for the purpose of this academic project, the system can run locally on XAMPP or be hosted on a free web server. The premium subscription model can generate revenue in a real-world scenario, making it economically viable.

Cost Estimates:

1. Domain (for deployment): 2,000 PKR per year
2. Hosting (shared hosting): 10,000 PKR per year
3. Development cost: 0 PKR (because the system is developed by the student)
4. Total estimated cost: 12,000 PKR per year

Benefits Estimates

The Rishta System provides several important benefits. It saves time by allowing users to search for matches online instead of traditional methods. It reduces manual work because all records are stored automatically in the database. The system also improves privacy, security, and communication between users. In the future, the platform can also generate revenue through premium subscriptions.

1.2.3 Schedule Feasibility

A realistic schedule was prepared using the Gantt chart, considering the academic timeline. All phases are well-defined, and milestones are achievable within the given period. Regular meetings with the supervisor ensure that the project stays on track.

1.2.4 Specification Feasibility

The requirements are clearly defined and within the scope of a BIS final year project. Each functional requirement can be implemented using the chosen technologies. There is no requirement that is technically impossible or too complex.

1.2.5 Motivational Feasibility

The project addresses a real social need, which keeps the development team motivated. The supervisor’s guidance and the importance of the project for the degree also drive motivation

1.2.6 Legal & Ethical Feasibility

All data collected is with user consent. The system complies with data protection principles (not sharing data without permission). Photos are watermarked to prevent unauthorized use. The platform discourages fraudulent activities through verification. No copyrighted code or assets are used without permission.

Chapter 2

Problem Defining

The existing matchmaking methods face several critical issues that this project aims to solve. Traditional systems rely mainly on family networks, word-of-mouth, and newspaper advertisements, which are time-consuming and limited in reach. These methods also lack proper privacy and security, as personal information and photographs are often shared without user control.

Another major problem is the difficulty in filtering suitable matches based on preferences such as age, education, profession, and city. Communication between potential matches is also not structured, which creates misunderstandings and delays. In addition, manual record management makes the process inefficient and difficult to maintain.

The Rishta System is developed to overcome these problems by providing a secure, efficient, and user-friendly digital platform for matchmaking.

2.1 Problem of class timetable

Right now, University of Sahiwal is using the criteria of pasting the timetable on the classes and office bulletin board and students have to approach to the board to check their relevant class timetable. This is also a hectic for a student and even for a teacher. Because sometimes university management has to change the timetable accordingly to the shortage or unavailability of teacher. Or sometimes due to sudden shutdown of university in response of city situation. At that time most of the students and teachers don't get the new timetable on time and most of the students miss their classes. It's not easy for each and every student in person to go and check the bulletin board every time if the timetable has been changed or still not work upon it.

2.2 problem of Reach and Access

Traditional matchmaking systems mainly rely on personal networks, relatives, friends, and newspaper advertisements, which have very limited reach. Most individuals are restricted to their immediate social circle, which reduces the chances of finding a suitable life partner who meets all the required criteria such as education, profession, age, and location. If a suitable match is not available within the family or community circle, the process becomes slow and difficult.

In addition, traditional methods do not effectively use modern technology or the internet to expand search options. People from different cities or backgrounds cannot easily connect with each other, which further limits opportunities. As a result, many individuals spend a long time searching for a suitable partner due to the limited reach and accessibility of traditional matchmaking methods.

2.3 Problem of Privacy

In conventional methods, personal details (photos, phone numbers, family background) are shared with multiple intermediaries, leading to privacy breaches. Users have no control over who sees their information, and once shared, it cannot be retracted.

2.4 Problem of Authenticity

It is extremely difficult to verify the information provided by potential matches. Misrepresentation of age, profession, marital status, or even identity is common. This lack of trust discourages genuine individuals

2.5 Problem of Inefficient Filtering

Manually filtering matches based on dozens of criteria (age, education, city, etc.) is tedious and time-consuming. There is no systematic way to shortlist candidates who match one's specific preferences.

2.6 Communication Problem

Initiating a conversation can be awkward and may cause embarrassment. There is no formal, safe channel to express interest without fear of rejection or misinterpretation. Often, communication is delayed or lost

2.7 Security and Safety

Meeting strangers through ads or referrals poses safety risks. There is no mechanism to report or block malicious individuals. Users are vulnerable to harassment or fraud.

2.8 Manual Administration

It is tough to update timetable when classes are already allocated or allotted to teachers. Sometimes at the same time classes are mismanaged and both classes are there out of room to attend the lecture so finding another room is totally problematic at that time.

2.9 Subscription and Access

Without a digital system, there is no clear distinction between basic and premium services. Users have no option to pay for enhanced features, and the service provider cannot monetize effectively.

2.10 Lack of Data-Driven Insights

Manual systems provide no analytics. There is no way to understand user preferences, match success rates, or popular demographics, making it impossible to improve services.

Chapter 3

Software Requirement Specification

This chapter will discuss the detail of necessary system requirements required to develop the system. It will elaborate the functional requirements and non-functional requirements.

1.1 Introduction

1.1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of the functional and non-functional requirements of the Rishta System. It is intended to guide developers and ensure that the final product meets user expectations.

1.1.2 Document Convention

1. Bold text indicates key terms.
2. Monospaced text is used for code, variables, and file names.
3. Italics are used for emphasis.

1.1.3 Intended Audience

This document is intended for multiple audiences involved in the development and evaluation of the Rishta System. It provides project developers with a clear understanding of the system requirements, design specifications, and implementation guidelines, enabling them to develop and maintain the platform efficiently. It assists the project supervisor in monitoring progress, reviewing design choices, and providing constructive feedback throughout the development process. Additionally, the document serves as a reference for the evaluation committee, allowing them to assess the objectives, methodology, functionality, and overall effectiveness of the system during the final project evaluation. By addressing the needs of these audiences, the document ensures that all stakeholders have a comprehensive view of the project and its deliverables.

1.1.4 Project Scope

The scope includes:

1. User registration and authentication
2. Profile creation and management
3. Advanced search and matchmaking
4. Interest request system
5. Chat system (mutual interest based)
6. Admin dashboard with user management, verification, reports
7. Subscription management
8. Responsive web interface

Out of scope: Mobile apps (future work), payment gateway integration (simulated for academic purposes).

1.2 Overall Description

1.2.1 Project perspective

The Rishta System is a new, self-contained web application designed to provide a modern alternative to traditional matchmaking methods. It does not replace any existing system but aims to enhance accessibility and efficiency by leveraging online technologies, offering users a more convenient way to find compatible life partners.

1.2.2 Project Function

The system encompasses several key functions to support both users and administrators. It provides user management to handle registration, authentication, and account settings. Profile management allows users to create, update, and maintain detailed personal profiles. Search and matching functionality enable users to find suitable partners based on specified criteria, while interest management allows expressing and tracking interest between users. The platform also facilitates communication through chat and notifications, offers admin management for controlling users and content, and supports reporting for monitoring system activity and generating insights.

1.2.3 User Classes and Characteristics

The system serves multiple classes of users. Guests (visitors) can view limited preview profiles and are encouraged to register for full access. Registered users (free) can create profiles, search for matches, send interests, and view a limited number of profiles, while chat functionality remains restricted. Premium users enjoy unlimited profile views, full photo visibility, active chat capabilities, and priority placement in search results. The administrator has full access to all data, including user management, reports, and system settings, ensuring smooth operation and oversight.

1.2.4 Operating Environment (OE)

The Rishta System is hosted on a server environment supporting Apache, PHP, and MySQL, enabling users to access the application through web browsers on any device with an internet connection. This setup ensures platform independence and accessibility across desktops, tablets, and mobile devices.

1.2.5 Design and Implementation Constraints (DCI)

The design of the system must be intuitive and user-friendly for all age groups, with a responsive interface compatible with mobile browsers. Implementation is constrained to the use of PHP and MySQL, and no paid third-party services are to be used, ensuring cost-effectiveness and maintainability.

1.3 User Documentation

Comprehensive documentation will accompany the system. A user manual (PDF) guides users through registration, profile setup, search, sending interests, and chat features. An admin manual provides instructions for managing users, verifying profiles, and generating reports. Additionally, an online FAQ and help section will be integrated within the application to support users with common issues.

1.4 Assumptions and Dependencies (AD)

The system assumes that users possess basic computer and internet skills and provide accurate personal information. Key dependencies include properly installed and configured Apache, PHP, and MySQL on the server

1.5 External Interface Requirements

1.5.1 User Interfaces

The user interface is designed to be clean and modern, utilizing **Bootstrap** for responsive layouts that adapt to different screen sizes. Forms include real-time validation for enhanced user experience, while the admin interface leverages **data tables** for efficient management and navigation of system data.

1.5.2 Software Interfaces

The application uses PHP scripts to handle HTTP requests and communicates with the MySQL database via MySQL or PDO. The system also interacts with the server file system to store and retrieve profile photos securely.

1.5.3 Communication Interfaces

Communication is primarily conducted over HTTP/HTTPS protocols, with AJAX (JavaScript) employed for asynchronous operations such as chat messaging and interest updates. Optional email notifications are handled through the PHP mail function to keep users informed of relevant activities.

1.6 Project Features

The proposed application has some important and basic features which are necessary. Profile of the application consists of the following features.

1.6.1 Profile Privacy & Photo Security

We help you save time to focus on what's important. A timetable management system for businesses to efficiently measure the time spent on projects, meetings, interviews. Know where your time goes. It is especially required for the academic sessions because when there are a lot of students studying at the same place it is hard to show them a paper description of timetable. Some of them go to noticeboards somehow timetable changes due to some reasons they don't even know but they follow the old one.

1.6.1.1. Functional Requirements

1. Users can set photo visibility: public, only matched, only premium.
2. All uploaded photos are watermarked with a semi-transparent logo using PHP GD library.
3. Watermarking happens at upload time; original file is not stored.
4. Profile photos are stored in a folder with random filenames.

1.6.1.2. Non-Functional Requirements:

1. Watermark must be clearly visible but not intrusive.
2. Access control must be enforced at both PHP and database level.

1.6.2 Search & Matching Logic

1.6.2.1. Functional Requirements

1. Search filters: age range, city, education, religion, profession, income range (optional).
2. Results sorted by compatibility score (calculated based on matching profile fields and preferences).
3. Compatibility score stored temporarily or calculated on the fly.
4. Users can save searches (future enhancement).

1.6.2.2. Non-Functional Requirements

1. Search queries must be optimized with MySQL indexing.
2. Response time under 3 seconds for up to 10,000 profiles.

.

1.6.3 Interest Requests & Chat

1.6.3.1. Functional Requirement

1. Send interest with a short optional message.
2. Interest status: pending, accepted, rejected.
3. Upon acceptance, both users can see each other's hidden contact info (phone, email) and chat is enabled.
4. Chat interface shows message history, allows reporting of individual messages.
5. Admin can view reported messages and suspend users.

1.6.3.2.

2. Send interest with a short optional message.
3. Interest status: pending, accepted, rejected.
4. Upon acceptance, both users can see each other's hidden contact info (phone, email) and chat is enabled.
5. Chat interface shows message history, allows reporting of individual messages.
6. Admin can view reported messages and suspend users.

6.1 Other Non-functional features

6.1.1 Performance Requirements

1. Page load time ≤ 3 seconds on average.
2. Concurrent users: at least 50.

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6.1.2 Safety Requirements

1. Regular database backups.
2. Error logging for debugging.
3. Graceful handling of server errors (custom error pages).

6.2 Security Requirements

1. Passwords hashed using password hash ().
2. Prepared statements or parameterized queries to prevent SQL injection.
3. Output encoding with html special chars () to prevent XSS.
4. Session fixation protection by regenerating session ID after login.
5. Admin pages protected by session and role check.

6.3 Software Quality Attributes (SQA)

6.3.1 Correctness

All features must work as specified. The compatibility score must be computed accurately. Interest status transitions must follow the defined logic.software system.

6.3.2 Reliability

The system should be available 99% of the time during testing. Database connections must be closed properly.

The blunder rate consists on the occurrence of given contributions and on the possibility that an input will lead to solution an error.

6.3.3 Suitability

The interface should be intuitive, with clear labels, tooltips, and error messages

6.3.4 Maintainability

Code should be modular (separate files for functions, configuration, database connection). Comments and documentation provided.

6.3.5 Portability

The system can be deployed on any server with PHP 7.4+ and MySQL 5.7+.

6.4 Business Rules

1. A user must be at least 18 years old to register.
2. Premium subscription duration: 1 month, 3 months, 6 months, 1 year (defined by admin).
3. Free users can view up to 10 profiles per day (configurable).
4. A user cannot send more than 20 interests per day (to prevent spam)

Chapter 4

Methodology

The project followed an iterative and incremental approach, combining elements of the waterfall model for planning and prototyping for user feedback. The process consisted of:

Requirement Gathering: Interviews with potential users, analysis of existing matchmaking platforms.

System Design: Database design, UI wireframes, system architecture.

Implementation: Coding in phases: basic authentication, profile management, search, interests, chat, admin panel.

Testing: Unit testing of PHP functions, integration testing of modules, UAT with a small user group.

Deployment: Local deployment using XAMPP for final demonstration

4.1 Tools

Tool	Purpose
VS Code	Code editor with PHP, HTML, CSS, JS extensions
XAMPP	Local server (Apache, MySQL, phpMyAdmin)
phpMyAdmin	Database management
Git	Version control (optional)
Chrome DevTools	Debugging frontend

4.2 Software Specifications

Operating System	Windows 10 Pro
XAMPP	3.3.0 (Apache 2.4, MySQL 8.0, PHP 7.4)
Code Editor	Antigravity

Table 3 Software Specifications

4.3 Hardware Specifications

Processor	Intel Core i5-8250U
RAM	1GB
Hard disk drive	16GB

Table 4 Hardware Specification

Chapter 5

Detailed Design & Architecture

5.1 Introduction

This chapter describes the architectural design, database schema, and detailed module design of the Rishta System.

5.2 System Architecture

The system is designed using a three-tier architecture, which separates the application into distinct layers, each with specific responsibilities. The presentation tier is the front-end interface that users interact with through their web browsers, utilizing HTML for structure, CSS for styling, Bootstrap for responsive design, and JavaScript for interactivity. The logic tier, or business logic layer, consists of PHP scripts running on the server that handle user requests, enforce business rules such as matching criteria and chat permissions, and manage communication with the database. The data tier serves as the back-end storage, where the MySQL database stores all persistent information including user profiles, messages, and administrative data. Requests from the client are processed by the logic tier, which retrieves or updates data from the database and sends the results back to the presentation tier for display. This layered approach ensures a clear separation of concerns, making the system more organized, maintainable, and scalable.

5.2.1 Architecture Design & Approach

The architecture of the Online Rishta Management System is a pre-planned arrangement of application features that defines the core functionality for users and guides the development lifecycle. This system is designed to ensure that the requirements of users are fully met. The application is intended for Android smartphones, providing a smart and responsive experience. Any errors or bugs found during usage will be promptly addressed. Continuous enhancement of modules will ensure stability and maintain the system's relevance in the competitive market.

5.2.1 Architecture Design

The architecture of the application is a crucial element that directly impacts user engagement. A well-structured and visually appealing architecture attracts users and encourages regular interaction. In the Rishta Management System, the architecture is designed to allow users to register, manage their profiles, search for matches, communicate with other users, and provide feedback. The main modules, including Profile Management, Matchmaking, Communication, and Feedback & Monitoring Systems, are integrated into a single cohesive model. This structure ensures smooth navigation, timely notifications, and an overall seamless experience while maintaining proper interrelationships between all components of the application.

5.2.2 Subsystem Architecture

The subsystem architecture divides the application's functionalities into smaller, independent modules that collectively provide the full set of features. In the Rishta Management System, the primary modules include Profile Management, Matchmaking, Communication, and Feedback & Monitoring. The Profile Management module allows users to create and update their personal profiles, upload photos, and manage privacy settings. The Matchmaking module enables users to search for potential matches, receive suggested matches, and send or respond to interest requests. The Communication module facilitates chat and notifications between users, ensuring clear and effective interaction. Finally, the Feedback & Monitoring module allows users to provide feedback, while administrators monitor activities, detect misuse, and maintain system integrity. Each subsystem ensures continuity, efficiency, and data security within the overall system.

5.3 Detailed System Design

The system design defines the workflow of the application, providing an organized and interactive experience for users. Upon launching the app, users are directed to a login or registration screen. Existing users can log in using their credentials, while new users are guided through the registration process. After authentication, users access the home screen, which displays the various features available within the app, such as profile search, sending interest, chatting, upgrading subscriptions, and providing feedback. The system ensures proper integration between modules, allowing smooth navigation and interaction between different functionalities.

5.3.1 Classification

The main features of the Rishta Management System include the Profile Management System, Matchmaking System, Communication System, and Feedback & Monitoring System. Each feature is designed to handle specific tasks and operates with backend support for data storage and processing. Profile Management stores user information securely, Matchmaking handles search and suggested matches, Communication enables interaction, and Feedback & Monitoring allows administrators to oversee activities and gather user insights.

5.3.2 Definition

The Profile Management module allows users to manage personal information, photos, and privacy settings. The Matchmaking module provides consistent search results, suggested matches, and interest management, helping users find compatible profiles efficiently. The Communication module ensures smooth interaction between users, delivering messages and notifications in a clear and organized manner. The Feedback & Monitoring module enables users to submit feedback regarding the app or other users, while administrators monitor activities to maintain quality, security, and system effectiveness.teachers are present or not in the institute.

5.3.3 Responsibilities

5.3.4

The Profile Management system ensures users can create and update profiles securely, maintaining privacy. The Matchmaking system provides intelligent search, suggested matches, and interest handling. The Communication system allows users to chat and receive notifications, reducing misunderstandings or delays. The Feedback & Monitoring system collects user feedback and tracks activities, enabling administrators to detect issues, monitor performance, and implement improvement.

Use case Diagram

Figure 1Admin Use Case Diagram

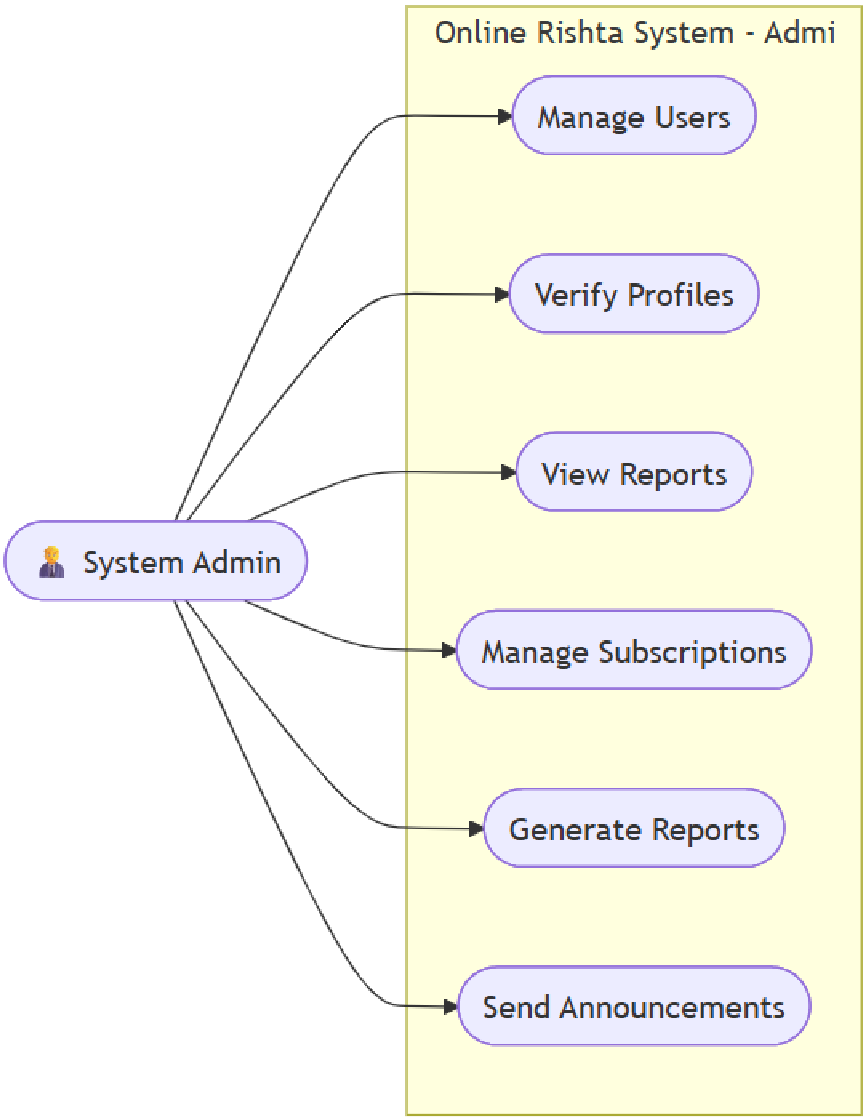
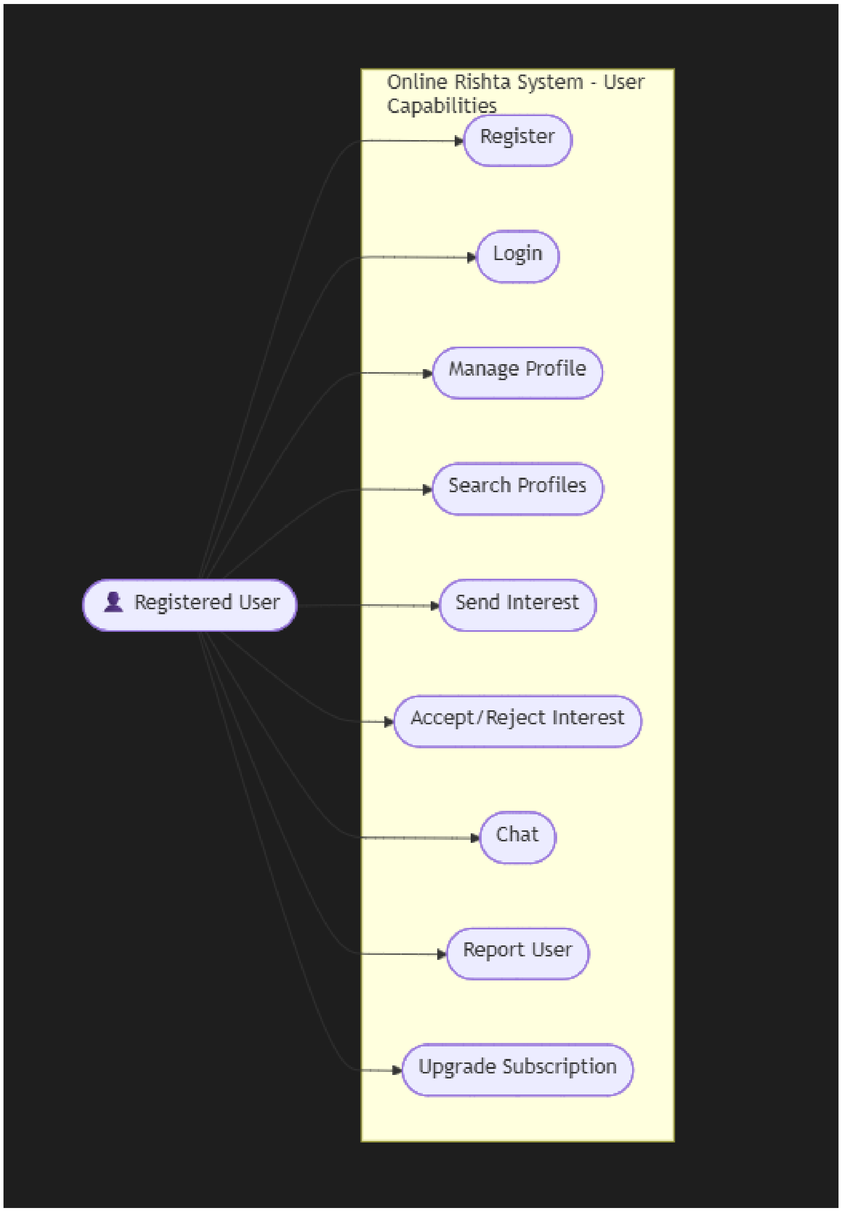


Figure 2 : user Use Case Diagram



5.3.10.3 Sequence Diagram

A sequence diagram describes that how a group of objects work together. This diagram is used by software developers to understand requirements for new software. In this diagram groups are involved to upload documentation through this given process in Campus on mobile.

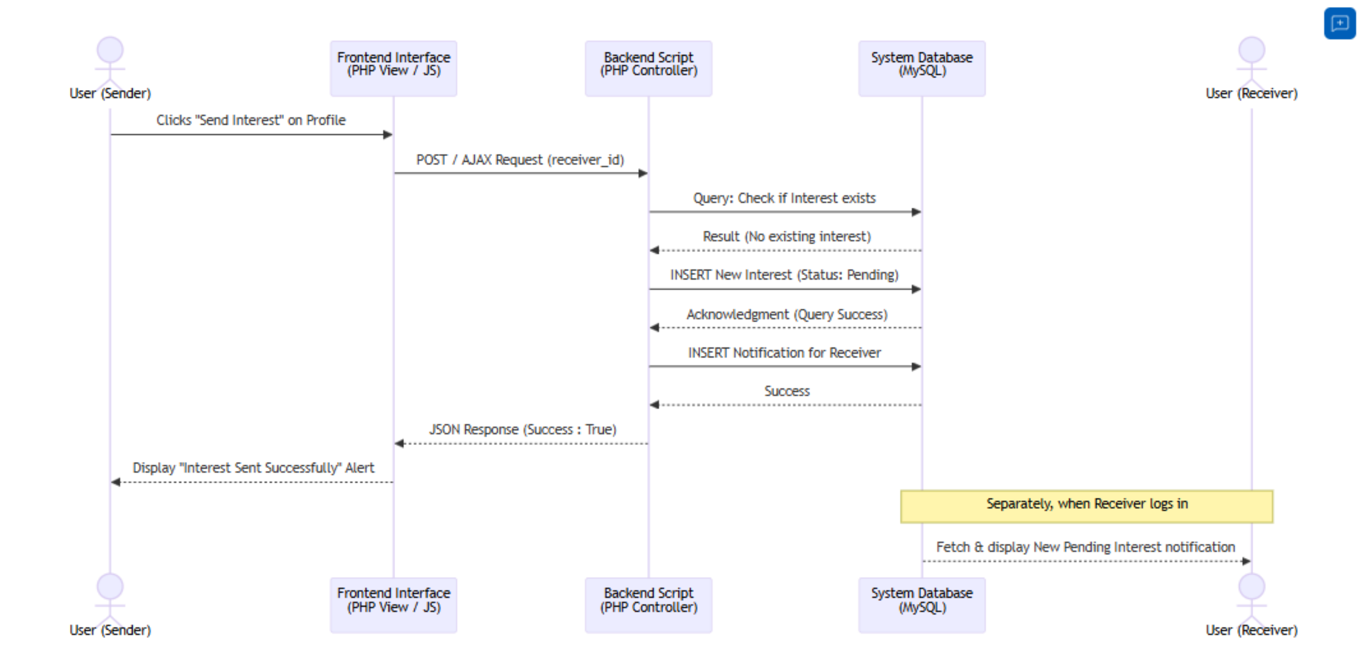


Figure 3 Sequence Diagram

Chapter 6

Implementation & Testing

6.1 Implementation Details

In this chapter we go through a light on testing plans and implementations.

6.1.1 Project Structure

```
text
rishta-system/
├── admin/
│   ├── css/
│   ├── js/
│   ├── includes/
│   ├── process/
│   ├── index.php
│   ├── dashboard.php
│   ├── users.php
│   ├── user_management.php
│   ├── reports.php
│   ├── settings.php
│   ├── subscriptions.php
│   ├── support.php
│   ├── messaging_system.php
│   ├── announcements.php
│   ├── payments.php
│   ├── profile_management.php
│   ├── verifications.php
│   ├── activity_logs.php
│   ├── matches.php
│   └── user_details.php
│
├── user/
```

```
│   ├── api/
│   ├── process/
│   ├── dashboard.php
│   ├── profile.php
│   ├── view_profile.php
│   ├── search.php
│   ├── matches.php
│   ├── interests.php
│   ├── chat.php
│   ├── notifications.php
│   ├── preferences.php
│   ├── settings.php
│   ├── subscription.php
│   ├── payment_history.php
│   ├── payment_checkout.php
│   ├── shortlist.php
│   ├── visitors.php
│   └── activity.php
│
├── config/
│   ├── database.php
│   └── functions.php
│
├── includes/
│   ├── header.php
│   └── footer.php
│
├── assets/
│   ├── css/
│   ├── js/
│   └── images/
│
├── index.php
```

├── login.php

├── register.php

├── logout.php

├── readme.txt

└── database.sql

6.1.2 Key Implementation Features

- **Password Hashing:** password hash() with bcrypt.
- **Watermarking:** Uses image copy merge () from GD library.
- **AJAX Chat:** Polling every 5 seconds using jQuery.
- **Session Management:** session_start () and session regeneration on login.
- **Database Layer:** MySQL with prepared statements.

6.2 Testing

Testing was conducted at multiple levels:

6.2.1 Unit Testing

Individual PHP functions (e.g., calculate_compatibility(), watermark_image()) were tested with various inputs.

6.2.2 Integration Testing

Tested interactions between modules (e.g., interest acceptance triggers chat creation).

6.2.3 User Acceptance Testing

Five users (age 20-30) were invited to test the system and provide feedback. Issues like unclear error messages and slow search were fixed.

6.2.4 Security Testing

- SQL injection attempts using special characters in inputs were blocked by prepared statements.
- XSS attempts using <script> tags were neutralized by htmlspecialchars().
- Unauthorized access attempts to admin pages were redirected to login.

6.3 Test Cases

Test Case TC-01: User Registration

Field	Test Data	Expected Result
Full Name	"John Doe"	Valid
Full Name	"John123"	Error: Only letters

Field	Test Data	Expected Result
Email	"john@example.com"	Valid
Email	"invalid"	Error: Invalid email
Age	17 years	Error: Must be 18+
Password	"pass"	Error: Too short
Password	"Pass1234"	Valid (strength good)
Confirm Password	mismatch	Error: Do not match

Test Case TC-02: Login

Input	Expected Result
Valid email & password	Redirect to dashboard
Invalid email	Error message
Wrong password	Error message
Suspended account	Error: Account suspended

Test Case TC-03: Send Interest

Scenario	Expected Result
User A sends interest to B	Interest record inserted with status 'pending'
User B accepts	Status changes to 'accepted', chat unlocked
User B rejects	Status changes to 'rejected', chat remains locked

Test Case TC-04: Chat

Scenario	Expected Result
Mutual interest accepted	Chat interface visible for both
User A sends message	Message appears in database and visible to B via AJAX
User B reports message	Message flagged, admin notified

Test Case TC-05: Admin – User Management

Action	Expected Result
Search for user by email	Correct user displayed
Suspend user	User cannot log in
Delete user	User and all related data removed

Chapter7

Results & Discussions

7.1 User Interface Screenshots

(Insert screenshots with captions as listed in the List of Figures)

Figure 8: Registration Form

Shows fields with validation messages.

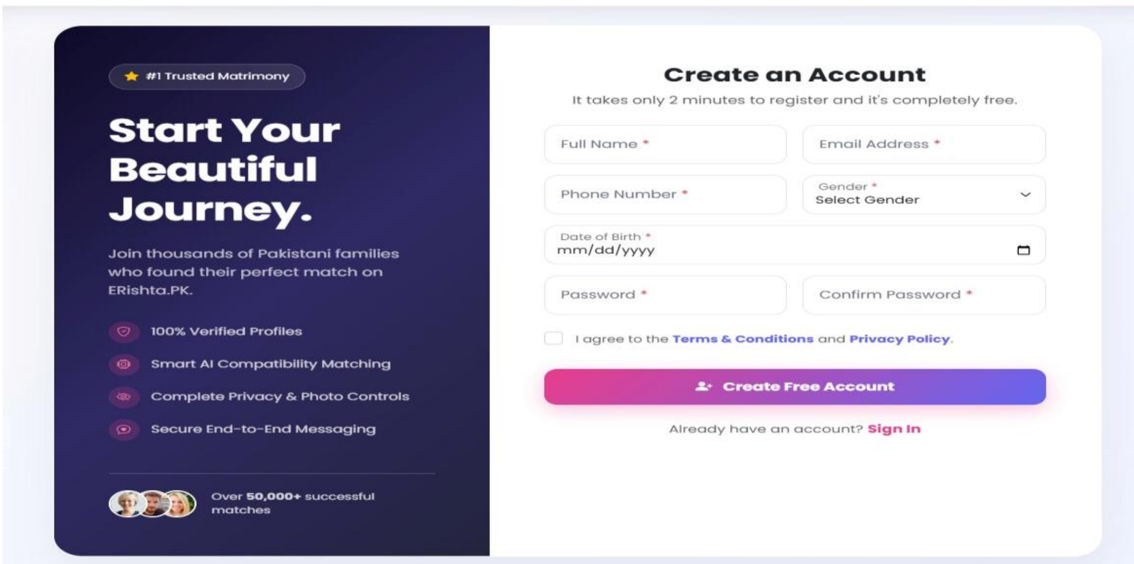


Figure 9: Login Page

Clean layout with email and password fields.

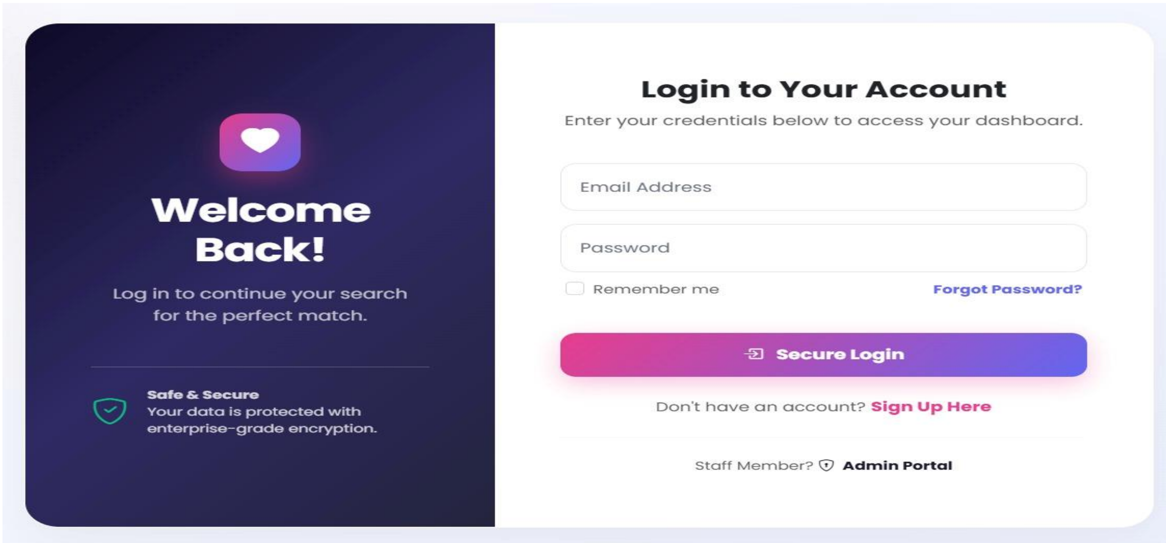


Figure 10: User Dashboard

Displays suggested matches, profile completion, recent activity.

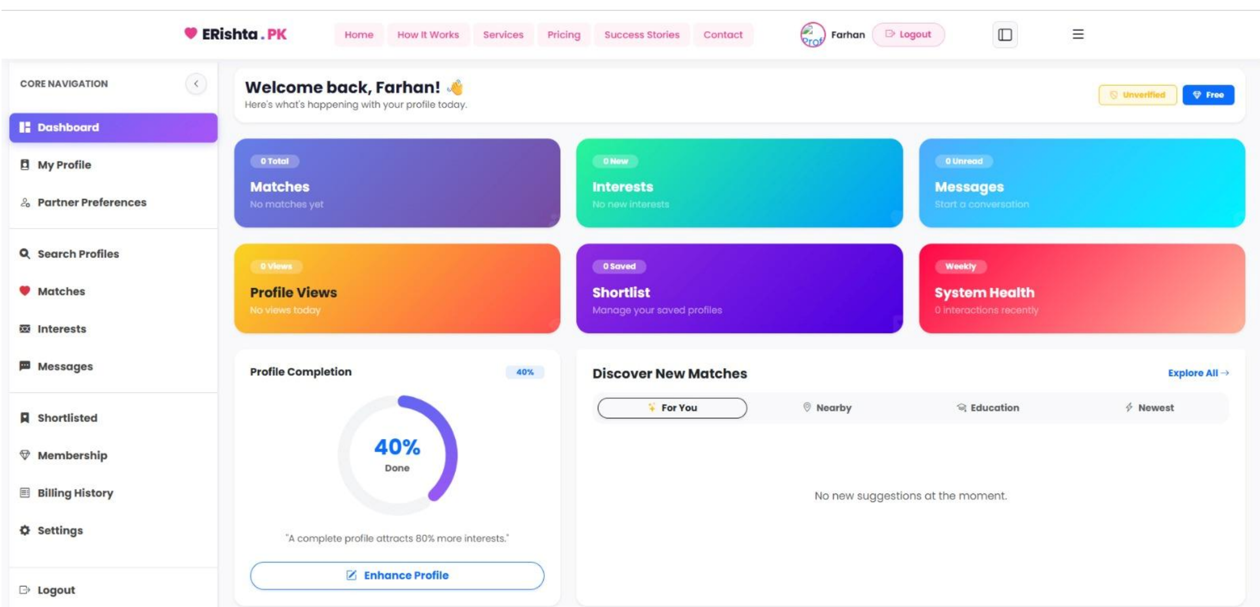


Figure 11: Search Results with Blurred Photos

For free users, photos are blurred; premium indicator shown.

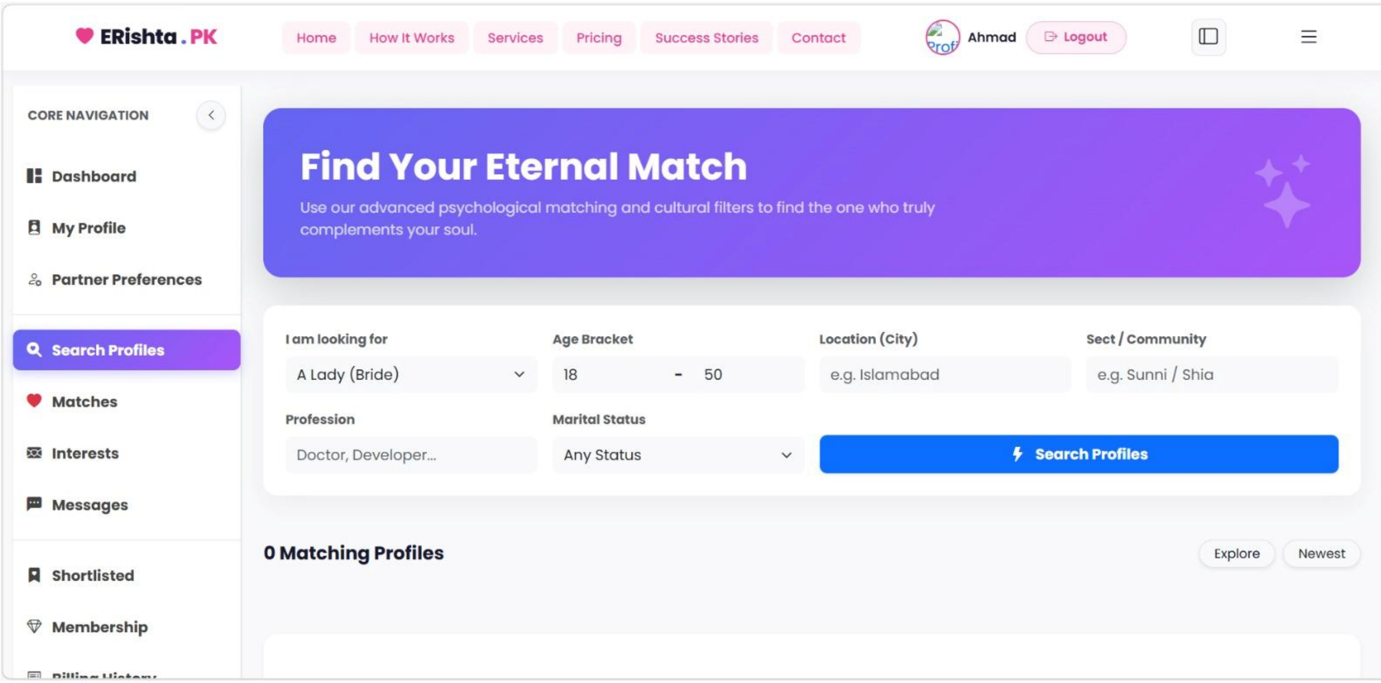


Figure 13: Interests Page

Sent and received interests with accept/reject buttons.

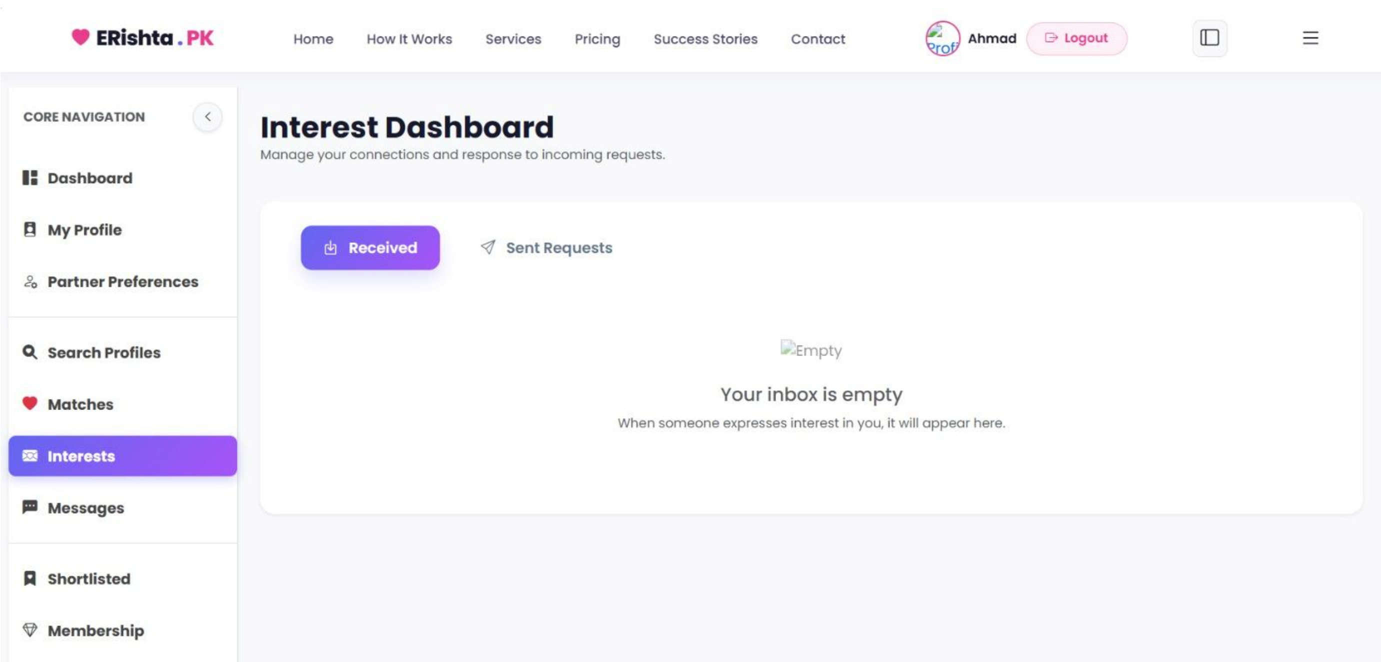


Figure 14: Chat Interface

Real-time-like chat using AJAX; report button for each message.

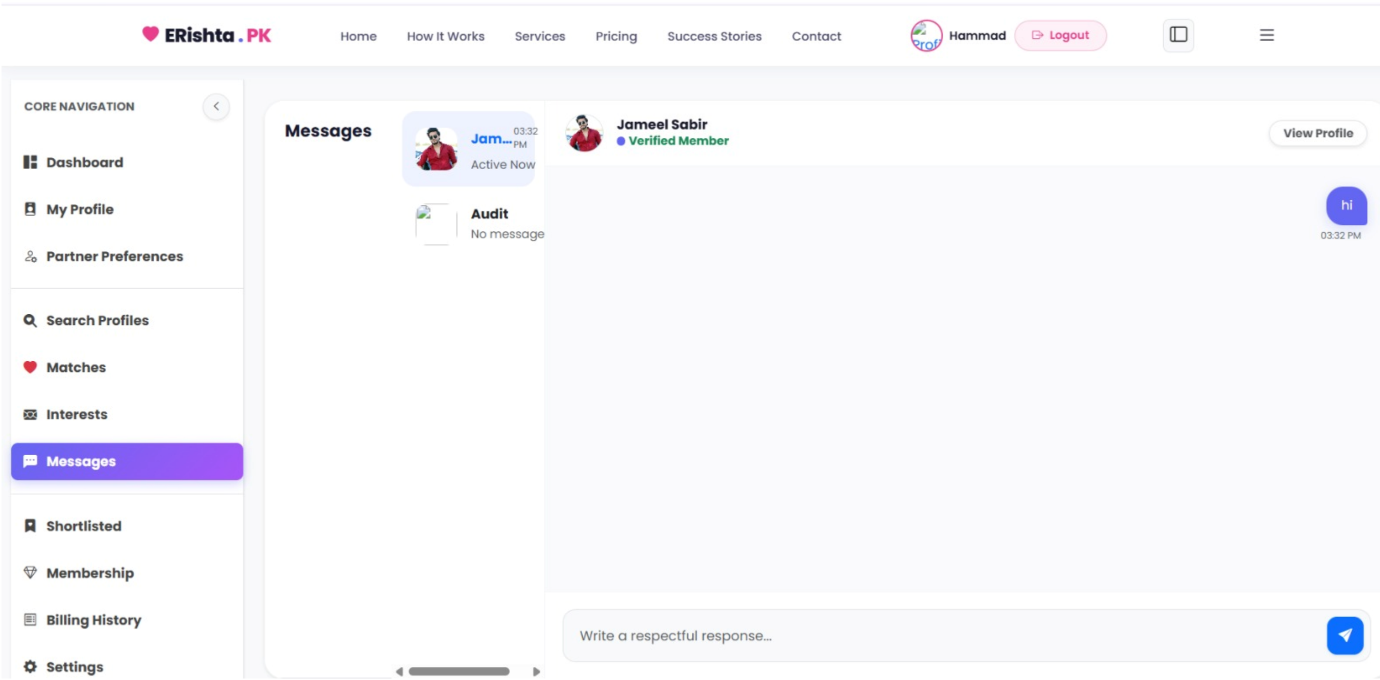


Figure 15: Admin Dashboard

Widgets showing total users, active users, pending verifications, etc.

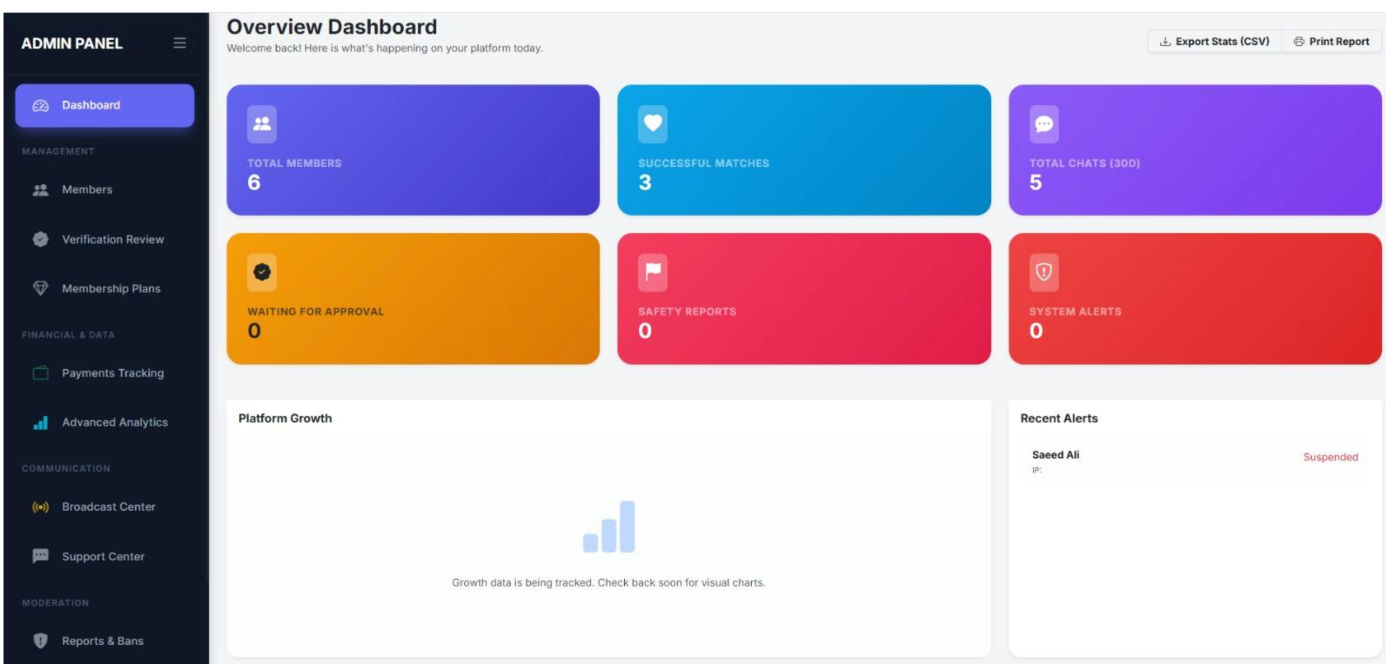


Figure 16: User Management

Table with user details, edit, suspend, delete options.

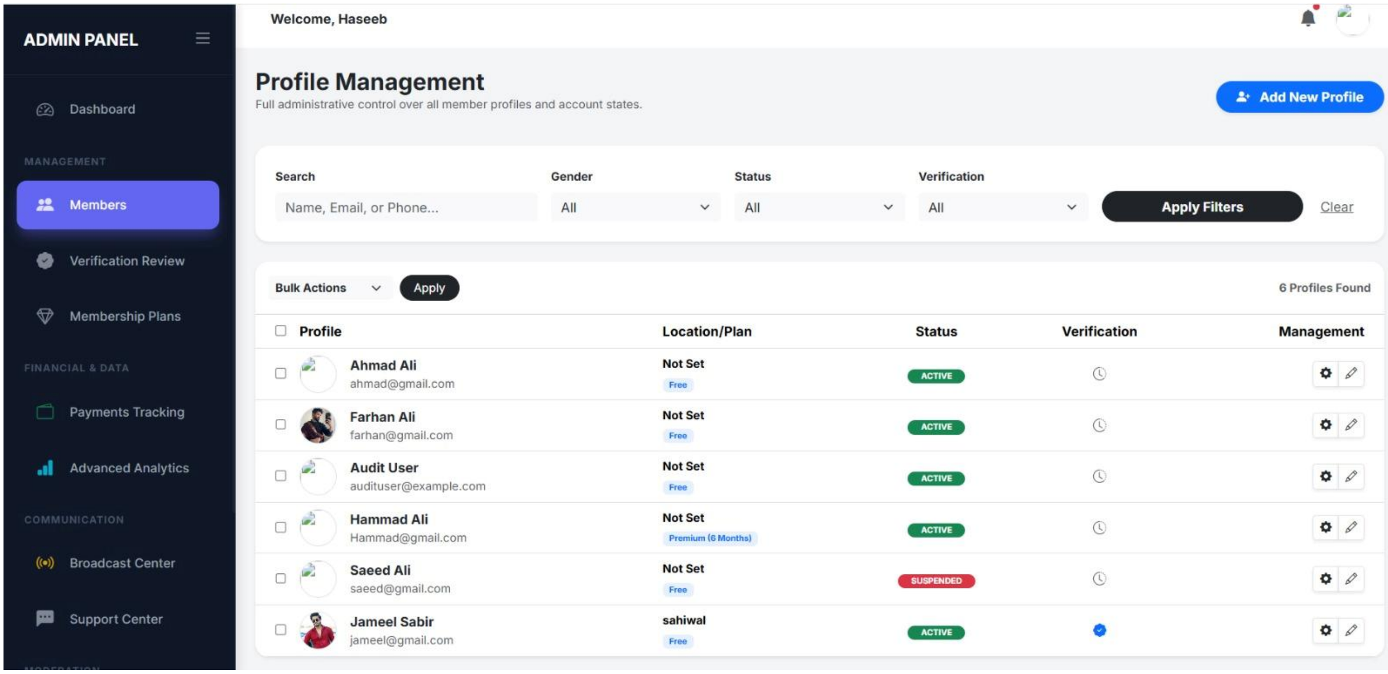


Figure 17: Profile Verification Queue

List of users awaiting verification with approve/reject buttons.

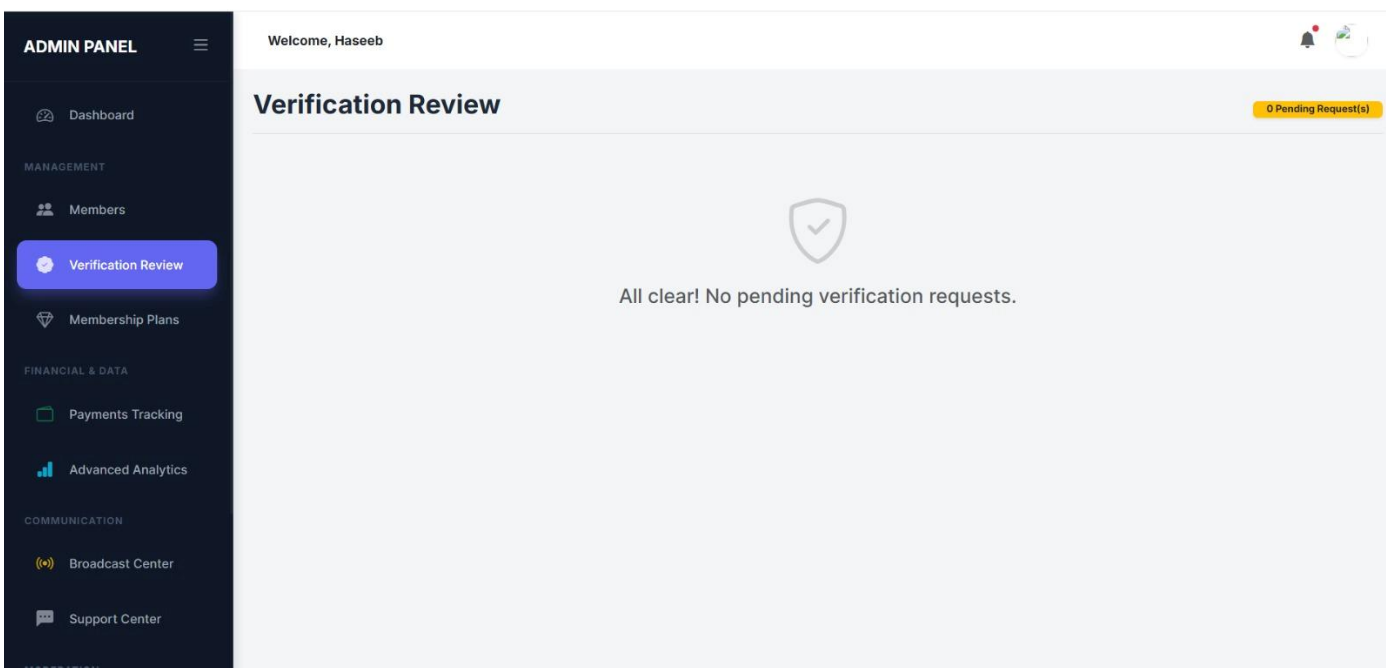


Figure 18: Message Moderation

Flagged messages with user details and action buttons.

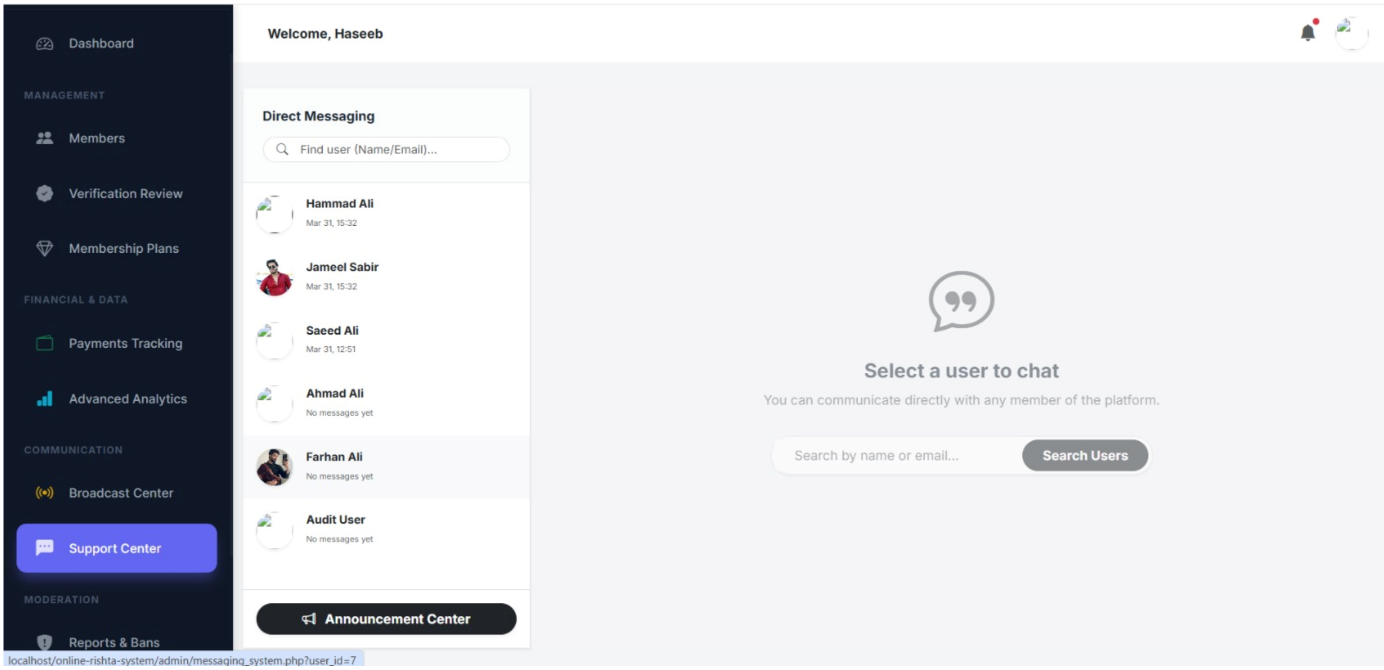


Figure 19: Subscription Plans

Admin view to add/edit plans.

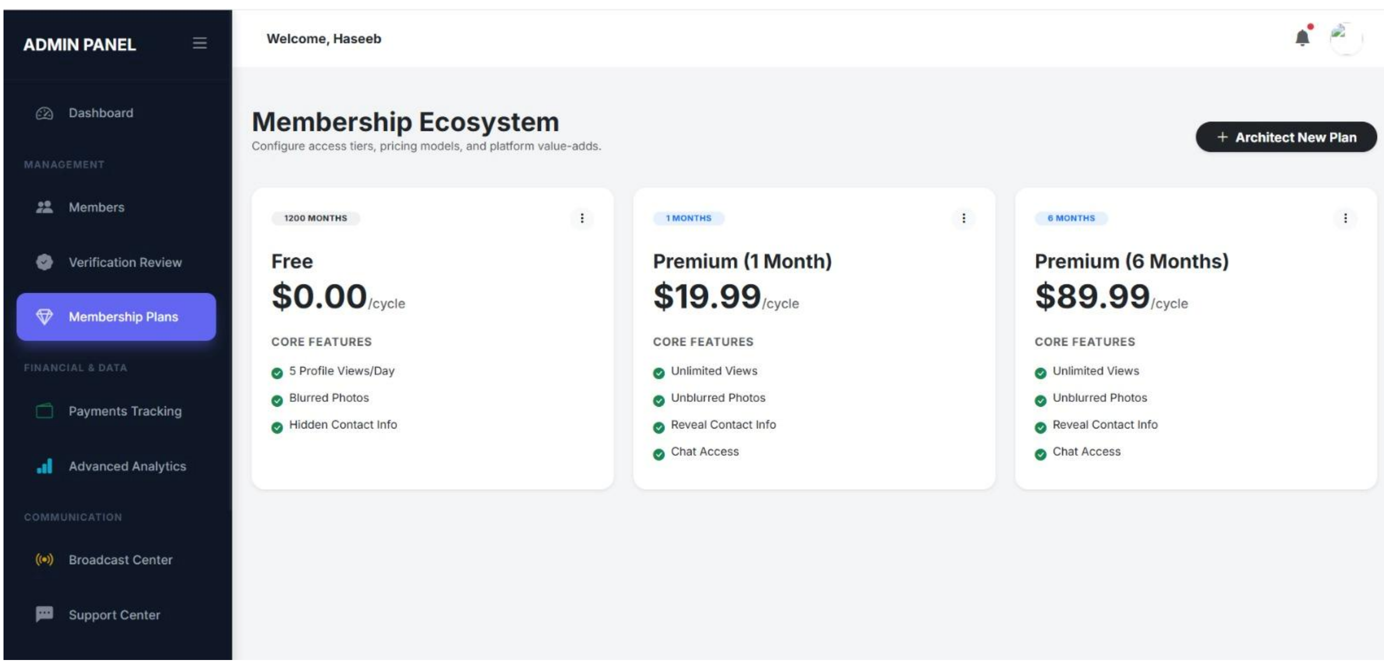
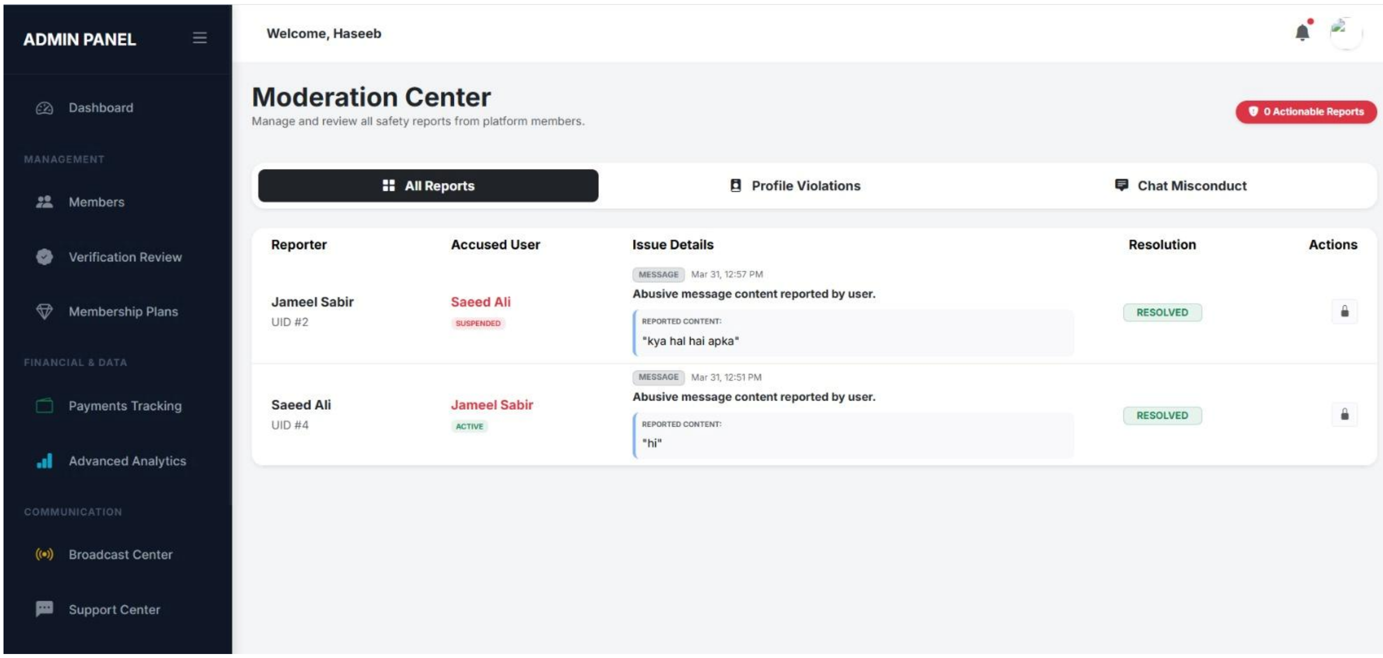


Figure 20: Reports Page

Date range picker to generate PDF/CSV reports.



7.2 Feature Verification

Feature	Status	Remarks
User Registration	Complete	Works with validation
Login/Logout	Complete	Session management
Profile Creation	Complete	Includes photo upload & watermark
Search & Filters	Complete	All filters functional
Compatibility Score	Complete	Score displayed
Interest Requests	Complete	Status tracking works
Chat	Complete	Works after mutual acceptance
Admin User Management	Complete	CRUD operations
Profile Verification	Complete	Admin approves/rejects
Reports	Complete	Basic reports generated

Feature	Status	Remarks
Subscription	Complete	Manual upgrade works

7.3 Challenges Faced

- **Image watermarking:** Initially, image quality was reduced; optimized by using appropriate compression.
- **AJAX chat polling:** Caused server load; mitigated by increasing poll interval and limiting active chats.
- **SQL injection prevention:** Adopted prepared statements throughout after initial prototyping with concatenated que

Chapter 8

Conclusion & Future Work

8.1 Conclusion & Scope

The Rishta System successfully provides a secure, feature-rich, and user-friendly digital matrimonial platform. It addresses the key issues of reach, privacy, authenticity, and communication. The use of open-source technologies (PHP, MySQL, HTML, CSS, Bootstrap, JavaScript) ensures that the system is cost-effective and easily deployable. The admin panel gives full control to manage the platform, while the user interface is designed to be intuitive. Testing confirmed that all core features work as intended, and feedback from users was positive.

8.2 Lesson Learnt

1. The importance of planning the database schema before coding.
2. Security must be integrated from the start, not as an afterthought.
3. Real-time features (chat) require careful handling of server resources.
4. User testing early in development can reveal usability issues.
5. Writing modular code simplifies testing and maintenance.

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Various online resources: Stack Overflow, MDN Web Docs, etc.